Baseline Methodology and Evidence Summary for X-ray Prohibited-Item Detection and Instance Segmentation

1. Purpose and Comparison Contract

This document summarizes the methods, published evidence, reproduction requirements, and experimental roles of the two selected baseline families. It deliberately omits line-by-line repository detail, which is placed in the companion *Architecture and GitHub Repository Guide*. Published values below are evidence from the cited papers; they are not results from our implementation.

The baselines are complementary rather than directly interchangeable. The 2023 PIDray journal model addresses the realistic mixture of positive and normal baggage images and performance on Easy, Hard, and deliberately Hidden items. The 2025 bilayer model addresses object-to-object overlap using additional occlusion-area supervision and SAM-based mask decoding. A numerical comparison is valid only within the same dataset version, annotations, split, input, evaluator, and supervision level.

Factor	**PIDray journal baseline**	**Occlusion-Aware Bilayer + SAM**
Primary protocol	Original PIDray	PIDray-A and PIXray-A
Supervision	Image status, boxes, instance masks; includes normal images	Boxes, target masks, and extra occlusion-area masks
Main mechanism	Image-level screening plus dense multi-scale features and Cascade Mask R-CNN	SAM/RSPrompter proposal-and-prompt pipeline with two mask decoders
Primary strength	Background-only images and Hidden evaluation	Explicit overlap reasoning and stronger mask decoding
Critical limitation	Hard R_0 gate can suppress positives; legacy framework	Extra annotations and large SAM ViT-L compute; weaker than PIDrayNet on Hidden
Role in our study	Required primary baseline	Conditional specialized baseline

2. Baseline 1: PIDray Journal Model

2.1 Correct model identity and data

PIDray contains 124,486 X-ray images across 12 prohibited-item categories. The journal release reports 47,677 images containing prohibited items and 76,809 normal images, with 76,913 training images and test subsets of 24,758 Easy, 9,746 Hard, and 13,069 Hidden images [1]. Positive instances have boxes and masks.

SDANet and the full journal model are not the same baseline. SDANet is the earlier selective dense-attention method retained in the journal comparison. The strongest 2023 system combines the journal's divide-and-conquer design with Cascade Mask R-CNN and is often called **PIDrayNet** by later work. Our primary reproduction target is the released full configuration, `configs/cascade_mask_rcnn_r101_with_R0R1.py`; SDANet remains a separately labeled published row.

2.2 Architecture and forward process

1. **ResNet-101 backbone.** Four stages produce C_2–C_5 with 256, 512, 1024, and 2048 channels at strides 4, 8, 16, and 32. The released setting uses ImageNet initialization, freezes the stem and first residual stage under MMDetection conventions, and evaluates BatchNorm statistics during training.
2. **Attentive feature pyramid.** A standard top-down FPN produces five 256-channel levels P_2–P_6. At every target level, the custom AttFPN resizes and aggregates all five levels. Channel weights select useful levels per channel; spatial weights select useful levels per position; an MMCV context block refines the fused representation; and a residual connection preserves the original level.
3. R_0 **coarse confidence node.** The public code sends the final enhanced level, normally P_6, through a normalized sine positional encoding and a one-query transformer decoder. A linear layer predicts whether the image contains any prohibited item. Binary cross-entropy is weighted by 0.1.
4. **Training routing.** Images with detached R_0 probability greater than 0.5 enter the detector branch. Detector gradients are scaled by the fraction of images selected in the current original batch. Normal images remain essential supervision for R_0 and for background-confusion analysis.
5. R_1 **Cascade Mask R-CNN.** An RPN proposes regions from five pyramid levels. Three box heads train at IoU thresholds 0.5, 0.6, and 0.7 with weights 1, 0.5, and 0.25. A four-convolution, class-specific mask head predicts 28×28 instance masks from 14×14 RoI features.

The conceptual paper presents R_0 as screening before R_1. In the released test function, however, the RPN and RoI heads run before low-confidence output scores are suppressed. Therefore the exact public implementation should not be credited with inference-time early-exit savings. The test gate also follows MMDetection's usual batch-one assumption.

2.3 Released training configuration

Item	Released / paper-aligned setting
Framework	Vendored MMDetection 2.25.0; MMCV 1.5.2; PyTorch 1.10; torchvision 0.11
Input	Aspect-ratio-preserving fit within 500×500; horizontal flip 0.5; ImageNet normalization; pad to multiple of 32
Optimization	SGD, learning rate 0.02, momentum 0.9, weight decay 10^{-4}; 500-iteration warm-up; drops at epochs 8 and 11; 12 epochs
Batch/hardware	Configuration: 4 images per GPU and 4 configured GPUs, effective batch 16; paper reports four RTX 3090 GPUs
Testing	Box and mask COCO evaluation; score threshold 0.05, box NMS 0.5, maximum 100 detections, mask threshold 0.5

Two details are methodologically decisive. First, the repository modifies `mmdet/datasets/custom.py` so that empty ground-truth images are not filtered. Importing a standard pip MMDetection instead can silently remove normal images. Second, the supplied configuration points validation and testing to the same test annotation. Our study must create validation only from the official training portion and reserve Easy/Hard/Hidden for final testing.

2.4 Published results and interpretation

Table 1: Published PIDray AP. "Overall" is the paper's combined result.

	Box AP				Mask AP			
Method	Easy	Hard	Hidden	Overall	Easy	Hard	Hidden	Overall
Mask R-CNN	66.2	58.6	43.8	59.1	59.2	50.1	35.5	51.2
Cascade Mask R-CNN	71.9	63.2	46.8	63.7	60.7	52.0	36.2	52.8
SDANet	72.5	63.7	48.0	64.4	61.1	51.7	37.0	52.9
PIDray journal model	**74.5**	**64.8**	**53.0**	**66.6**	**61.4**	51.9	**39.7**	**53.4**

Relative to Cascade Mask R-CNN, the full model gains 2.9 overall box AP and 6.2 Hidden box AP, but only 0.6 overall mask AP. This is central to our problem definition: feature fusion and screening substantially help localization, yet mask quality under overlap remains a separate weakness. The paper also shows that adding normal images sharply reduces image-level false alarms, which makes this baseline the strongest starting point for a background-confusion study.

3. Baseline 2: Occlusion-Aware Bilayer Modeling with SAM

3.1 Data and supervision

Ren et al. extend PIDray and PIXray with an occlusion mask for the area of another prohibited item overlapping each target instance, creating PIDray-A and PIXray-A [2]. The paper states that these areas were manually annotated. The repository also supplies `convert_occlusion_annotation.py`, which approximates extra masks from existing COCO annotations using overlap tests. Output from this converter must not be assumed identical to the released/manual "-A" annotations; its provenance and statistics must be reported separately.

The additional occluder mask changes the learning problem. A model trained on ordinary PIDray boxes and target masks does not receive equivalent supervision, so PIDray-A values cannot be placed in a same-protocol ranking with an original-PIDray run.

3.2 Repository lineage and architecture

The official Occ repository starts from the **RSPrompter** release code, which itself adapts SAM to automatic instance segmentation through learned prompts within a modified MMDetection framework. Occ vendors MMDetection 3.2.0 and MMPretrain 1.1.1, uses MMCV 2.1.0, imports the Hugging Face SAM ViT-L checkpoint, and uses PEFT LoRA adapters.

1. **SAM ViT-L representation.** With 512×512 input and patch size 16, the encoder produces a 32×32 embedding. ViT-L has width 1024, 24 transformer blocks, 16 heads, and global attention at blocks 5, 11, 17, and 23; a neck projects the output to 256 channels.
2. **What is frozen.** The paper describes a frozen SAM encoder. In the released configuration, the base encoder weights are frozen through PEFT while rank-16 LoRA adapters on the `qkv` projections are trainable (alpha 32, dropout 0.05). The two mask decoders are also trainable because `decoder_freeze=False`. Thus "base SAM encoder frozen except LoRA adapters" is the accurate code-level description.
3. **Five-level proposal features.** An identity feature aggregator passes the single SAM feature to a simple FPN, which creates five 256-channel maps at effective strides 4, 8, 16, 32, and 64. The RPN uses two anchor scales, 4 and 8, with ratios 0.5, 1, and 2.
4. **Boxes and learned prompts.** RoI Align provides fixed-size region features. A Shared-2FC box head predicts 12 PIDray classes and class-specific offsets. A prompt head converts positive RoI features into five SAM-compatible sparse prompt embeddings.
5. **Bilayer mask decoding.** The first SAM mask decoder predicts the occluding-object area. Its optimized image and point embeddings are added residually to the original embeddings. The second, separate SAM decoder then predicts the target mask under this guidance.

The supplied weight converter duplicates every original SAM `mask_decoder.*` parameter into `mask_decoder_bo.*`, so both decoder branches start from the same pretrained SAM initialization. The segmentation loss is

$$L_{\text{seg}} = L_{\text{target}} + 0.25 L_{\text{occluder}},$$

in addition to the RPN, classification, and box-regression losses.

3.3 Paper settings versus released configuration

Item	Paper	Released PIDray config
Backbone	Frozen SAM ViT-L	Base ViT-L frozen except LoRA adapters; trainable decoders
Input / optimizer	512×512; AdamW	Same family; AdamW learning rate 10^{-4}, weight decay 0.05, AMP
Batch / hardware	Batch 64 on one A800 80 GB	2 images per GPU, no declared gradient accumulation; repository tested 1 A800 or 8 RTX 4090 GPUs
Schedule	50 epochs; warm-up and cosine annealing	50 epochs; 50-iteration warm-up from factor 0.001; cosine minimum 10^{-7}
Annotations	Manually constructed PIDray-A	Reads `train_occ.json`/`test_occ.json`; converter available for custom COCO data

The paper/config batch discrepancy must be treated as a protocol difference, not silently reconciled. Exact reproduction should record the effective batch, learning-rate schedule, checkpoint source, and whether the official "-A" data or generated annotations are used.

3.4 Published results and interpretation

Table 2: Published overall PIDray-A results.

	Box			Mask		
Method	AP	AP50	AP75	AP	AP50	AP75
Mask R-CNN	58.7	76.2	67.2	50.2	73.4	58.2
Cascade Mask R-CNN	65.0	78.1	72.1	53.6	76.0	62.4
PIDrayNet	66.6	81.7	74.3	53.4	78.6	61.4
RSPrompter	66.0	83.1	74.3	56.5	81.4	64.4
Bilayer + SAM	**67.5**	**84.3**	**75.8**	**57.6**	**82.6**	**65.8**

Table 3: Published PIDray-A subset AP.

	Box AP			Mask AP		
Method	Easy	Hard	Hidden	Easy	Hard	Hidden
SDANet	72.5	63.7	48.0	61.1	51.7	37.0
PIDrayNet	74.5	64.8	**53.0**	61.4	51.9	**39.7**
RSPrompter	74.1	67.2	45.3	65.7	58.8	33.2
Bilayer + SAM	**75.7**	**68.3**	47.1	**67.1**	**60.0**	34.3

The bilayer model is strongest overall and on Easy/Hard in its protocol, but it does not beat PIDrayNet on Hidden. Its ablation moves from 66.1/56.6 box/mask AP with a simple SAM decoder, to 66.6/56.9 after adding the occluder decoder, and to 67.5/57.6 after adding residual guidance. The evidence therefore supports the bilayer design mainly as an overlap-aware segmentation baseline; it is not evidence that deliberate concealment has been solved.

4. Reproduction Methodology

Gate	Action	Output	Pass condition
G1	Pin sources and data	Commit IDs, package lock, dataset manifest, annotation checksums	Every artifact has fixed provenance.
G2	Audit loader/evaluator	Visual samples, empty-image counts, evaluator unit tests	Classes, masks, negatives, and subsets are correct.
G3	Evaluate official weights	Saved predictions and box/mask tables	Pipeline works before optimization is tested.
G4	Smoke train	One-batch and small-subset logs, gradient/loss checks	Intended branches receive finite gradients.
G5	Full reproduce	Frozen config, checkpoint, logs, metrics	Difference from the paper is quantified and explained.
G6	Diagnose failures	Ranked proposal, box, background, and mask errors	One actionable failure is identified.
G7	Compare improvement	Matched baseline, ablations, uncertainty and cost	Same data, supervision, evaluator, and tuning budget.

4.1 Baseline-specific checks

For PIDray, verify that normal images appear in training batches; log R_0 probability, positive-image gate recall, negative-image gate false-positive rate, selected fraction, and detector gradients; measure RPN recall before interpreting box/mask failure. For Bilayer + SAM, verify LoRA trainability, both decoder parameter groups, the target and occluder losses, transformed occlusion masks after flip/resize/crop, and correct duplication of the SAM decoder weights.

The released checkpoint should always be evaluated before a training run. If it fails to approach the published table, investigate annotation release, category IDs, data root, local-versus-pip package import, image preprocessing, checkpoint conversion, evaluator version, and subset construction before changing optimization.

5. Failure Analysis and Proposed-Method Connection

Background confusion should be measured at several levels rather than reduced to one image-level score:

Failure	Diagnostic	Possible intervention if dominant
Normal image receives detections	Alarm rate and detections per negative image; FP confidence and category	Hard-negative learning or calibrated image-level screening
Clutter region becomes a proposal	RPN objectness and unmatched proposal analysis	Background-aware feature/RPN refinement
Proposal is correct but class is wrong	RoI confusion matrix and score distributions	Class-aware hard negatives or representation separation
Hidden item has no proposal	Proposal recall by size and difficulty	Multi-scale/proposal feature improvement, not a mask-only module
Correct box, leaking/fragmented mask	Matched-mask IoU, boundary error, overlap-region audit	Boundary or occlusion-aware mask refinement

The provisional study component should target feature- or proposal-level background confusion because PIDray already contains an image-level R_0 gate. A defensible novelty claim must show that the new component addresses within-image clutter or hard proposals rather than merely reimplementing binary screening. The exact design will be selected after G6 and compared with the unchanged baseline through a minimal ablation matrix:

Run	Base system	Change	Question
A0	Cascade Mask R-CNN	None	Common anchor and diagnostic reference
A1	PIDray journal model	Official R_0 + AttFPN	Strong benchmark-aligned reference
A2	Matched selected base	Proposed component only	Does the component address the diagnosed error?
A3	Matched selected base	Full proposed method	Does the complete design improve the primary metric?
A4	A3	Remove/replace each essential term	Which mechanism causes the gain?

6. Required Reporting

Report box and mask AP/AP50/AP75, per-class and Easy/Hard/Hidden results, proposal recall, false positives per negative image, negative-image alarm rate, difficult-positive recall, and matched-mask overlap/boundary measures. Report parameters, peak memory, training time, batch-one latency, throughput, image size, numerical precision, and hardware. The primary comparison should include mean and standard deviation across seeds where feasible.

Every table must distinguish **published**, **reproduced**, and **proposed-method** values. Every run record must preserve the code commit, environment, configuration, dataset/annotation manifest, split seed, pretrained weights, optimizer, effective batch, random seed, checkpoint-selection rule, evaluator version, predictions, logs, and hardware. No threshold, module, or checkpoint may be selected on Easy, Hard, or Hidden.

7. Recommended Execution Order

1. Audit original PIDray and create the train-derived validation split.
2. Evaluate the released PIDray journal checkpoint, then reproduce Cascade Mask R-CNN and the full journal model.
3. Freeze the baseline error report and select the single proposed component.
4. Run same-protocol ablations and the final PIDray evaluation.
5. Only then obtain/audit PIDray-A, initialize both SAM decoders, and reproduce the bilayer track if compute permits.
6. Use PIXray/PIXray-A only as a separately documented generalization experiment.

This order keeps the primary background-confusion study independent of the heavier SAM experiment and prevents the specialized occlusion track from changing the main research question.

Primary Sources

[**1**] L. Zhang et al., "PIDray: A Large-Scale X-ray Benchmark for Real-World Prohibited Item Detection," *IJCV*, 2023: paper; official code.
[**2**] Y. Ren et al., "Prohibited Items Segmentation via Occlusion-Aware Bilayer Modeling," *ICME*, 2025: paper; official code.